

PART A [25 Marks]

This section consists of 3 questions. Answer **all** questions.

1 Evaluate the following limits:

a) $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$. [5 marks]

b) $\lim_{x \rightarrow -\infty} \frac{2x}{\sqrt{x^2 - 16}}$. [4 marks]

2 a) Find the derivative of $f(x) = x^2 + 2x$ using the first principle. [5 marks]

b) Find $\frac{dy}{dx}$ for each of the following

i) $y = \sin 4x + \ln(x^2 + 2)$. [2 marks]

ii) $y = e^{2x} \sec(3x + 1)$. [3 marks]

3 Given that $f(x) = 3x^4 - 4x^3 + 3$. Find the intervals where f is concave up and concave down. Hence find the inflection points. [6 marks]

PART B [75 Marks]

This section consists of 7 questions. Answer **all** questions.

- 1** Determine the value of a such that $\frac{\sqrt{3} + ai}{1 - \sqrt{3}i}$ is a real number and find this real number. [5 marks]
- 2** a) Solve $\sqrt{x+22} + \sqrt{x+6} = 8$. [5 marks]
b) Solve the inequality $\frac{4}{x+1} < x-2$. [6 marks]
- 3** a) Given $g(x) = x+3$ and $(f \circ g)(x) = x^2 + 1$, find $f(x)$. [3 marks]
b) The functions f and g are defined as $f(x) = 5-x$ and $g(x) = \frac{2x+3}{x-4}$,
 $x \neq 4$. If $(f \circ g^{-1})(a) = -10$, find the value of a . [5 marks]
- 4** Given a function $f(x) = e^{3x} + 2$.
a) Show that f is a one-to-one function. [2 marks]
b) Find the domain and range of f . [2 marks]
c) Determine the inverse function of f and state its domain and range. [5 marks]
d) Sketch the graphs of f and f^{-1} on the same axes. [3 marks]

5 a) Given

$$f(x) = \begin{cases} \frac{x^3 - 27}{x - 3} & , \quad x \neq 3 \\ 30 & , \quad x = 3 \end{cases}$$

i) Find $\lim_{x \rightarrow 3} f(x)$. [3 marks]

ii) Is f continuous at $x = 3$? Give your reason. [2 marks]

b) Determine the values of A and B such that the function

$$g(x) = \begin{cases} Ax + B & , \quad x \leq -2 \\ -x^2 + Ax - B & , \quad -2 < x < 2 \\ 4 & , \quad x \geq 2 \end{cases}$$

is continuous for all values of x . [6 marks]

c) Find the vertical asymptotes of $h(x) = \frac{2x}{\sqrt{x^2 - 16}}$. [3 marks]

6 a) Given

$$f(x) = \begin{cases} 2x^2 + 3 & , \quad x \leq 1 \\ 5x & , \quad x > 1 \end{cases}$$

Determine whether $f(x)$ is differentiable at $x = 1$ or not. [4 marks]

b) A curve is given by the parametric equations $x = \frac{t}{2t+1}$, $y = \frac{1}{(2t+1)^3}$.

Find $\frac{dy}{dx}$ in terms of t . [5 marks]

c) Given $y = e^x \ln(1+x)$, show that $(x+1)^2 \left(\frac{d^2y}{dx^2} - \frac{dy}{dx} \right) = xe^x$. [6 marks]

- 7 A cylindrical container has one opened and the other end closed. It has a circular base of radius r cm. Given that the total surface area of the container is $300\pi \text{ cm}^2$.
- a) Show that the volume of the container is $V = 150\pi r - \frac{\pi r^3}{2} \text{ cm}^3$. [4 marks]
- b) Find the maximum value of V . [6 marks]

END OF QUESTION PAPER

Final Answer:

PART A

1 a) $\frac{1}{4}$

b) -2

2 a) $2x + 2$

b i) $4\cos 4x + \frac{2x}{x^2 + 2}$

ii) $e^{2x} \sec(3x + 1)(3 \tan(3x + 1) + 2)$

3) C.U. : $(-\infty, 0) \cup \left(\frac{2}{3}, \infty\right)$, C.D. : $\left(0, \frac{2}{3}\right)$; I.P. : $(0, 3), \left(\frac{2}{3}, \frac{65}{27}\right)$

PART B

1) $a = -3$; $\sqrt{3}$

2 a) 3

b) $(-2, -1) \cup (3, \infty)$

3 a) $f(x) = x^2 - 6x + 10$

b) $a = 3$

4 b) $D_f = (-\infty, \infty)$, $R_f = (2, \infty)$

c) $f^{-1}(x) = \frac{1}{3} \ln(x - 2)$; $D_{f^{-1}} = (2, \infty)$, $R_{f^{-1}} = (-\infty, \infty)$

5 a i) 27

ii) discontinuous at $x = 3$

b) $A = 3$, $B = -2$

c) V.A. : $x = -4$, $x = 4$

6 a) not differentiable at $x = 1$

b) $-\frac{6}{(2t + 1)^2}$

7 b) $1000\pi \text{ cm}^3$